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# Artificial Intelligence Engineer

A career talk for senior secondary students. The pathways, the work, and the future of building intelligent systems.

## AUDIENCE

Classes IX to XII

## VENUE

KV RK Puram, Sector 2

## DEPARTMENT

DEPFE, NCERT, New Delhi

## AGENDA

# What does a career in AI really involve?

A structured walk through everything you need to decide whether this profession is right for you.

### 01 Educational Pathway

Two routes: degree or self taught

### 02 Nature of Work

Day to day tasks and tools

### 03 Working Conditions and Attributes

Environment and skills that help

### 04 Career Progression and Salary

Growth from entry to leadership

### 05 Opportunities and Colleges

Where to work and where to study

### 06 Courses and Future Outlook

Building employability for the long run

SECTION 01

# Getting into the field

There is no single fixed route. Whether through a degree or by building on your own, a strong grounding in mathematics and programming matters more than any one credential.

# 01

EDUCATIONAL PATHWAY

# Two ways into the field

There is no single fixed route. Pick the one that fits you, or blend both.

## ROUTE A

### The degree route

Structured, credential first

- 1 Science with Maths in Class XI and XII**  
Add Computer Science if offered, and start Python early.
- 2 Crack an entrance exam**  
JEE, BITSAT or CUET for IITs, NITs, IIITs and BITS.
- 3 B.Tech or B.Sc in CSE, AI-ML or Data Science**  
Optional M.Tech or PhD for research and senior roles.
- 4 Internships, then a full time job**  
Campus placements and referrals open the first role.

## ROUTE B

### The builder route

Self taught, portfolio first

- 1 Learn one language deeply**  
Master Python and the maths behind machine learning.
- 2 Build real projects**  
Ship them on GitHub and let the work speak for you.
- 3 Find clients and freelance**  
Take paid gigs, small contracts and open source work.
- 4 Improve, specialise, grow**  
Turn a real track record into full time and senior roles.

# What an AI engineer actually does

## Core Responsibilities

- Design, train and fine tune **machine learning and deep learning** models
- Collect, clean and prepare data, then engineer features
- Deploy models into products and **monitor them in production**
- Work on NLP, computer vision, recommendations and large language models
- Write reliable, production grade code and run experiments

## Tools and Technologies

- **Languages.** Python, SQL, sometimes C++ or Java
- **Frameworks.** PyTorch, TensorFlow, scikit learn, Hugging Face
- **Cloud and DevOps.** AWS, GCP, Azure, Docker, Git
- **Data.** Pandas, NumPy, Spark, vector databases
- Collaborates with data scientists, product and software teams

# The environment and what helps you succeed

## Hours and Setup

- Typical **40 to 45 hour** work weeks
- Extra hours near launches or deadlines
- **Hybrid and remote** work are common
- On site work for robotics and hardware

## Teams and Culture

- Small **cross functional teams**
- Agile sprints, code reviews, stand ups
- Global collaboration across time zones
- Constant need to keep learning

## Personal Attributes

- Strong **mathematics** and logical reasoning
- **Curiosity** and self driven learning
- Patience for experimentation and debugging
- Explaining complex ideas **simply**

SECTION 02

# Growth and reward

The path runs from intern to leadership. Compensation rises quickly with proven delivery and technical ownership.

# 02

PROGRESSION AND SALARY

# From entry level to leadership

## Intern or Trainee

Starting out

Learn on real projects, assist with data and experiments while completing studies.

## Junior AI or ML Engineer

0 to 2 years

Build and test models under guidance. Strengthen coding and data skills.

## AI or ML Engineer

2 to 5 years

Own features end to end, deploy models to production, mentor juniors.

## Senior AI Engineer

5 to 8 years

Lead complex systems, set technical direction, review architecture.

## Lead, Principal or ML Architect

8 to 12 years

Design large scale AI systems and guide multiple teams.

## Head of AI, Director or CTO

12 plus years

Own AI strategy and organisation wide impact.



ESTIMATED SALARY ALONG THE PATH

# Indicative compensation in India

|                                                |                                                  |                     |
|------------------------------------------------|--------------------------------------------------|---------------------|
| <b>Entry or Junior</b><br>0 to 2 years         | First roles. Varies widely by employer and city. | <b>6 to 12 LPA</b>  |
| <b>Mid level</b><br>2 to 5 years               | Proven delivery and production experience.       | <b>12 to 25 LPA</b> |
| <b>Senior</b><br>5 to 8 years                  | Technical ownership and mentoring.               | <b>25 to 45 LPA</b> |
| <b>Lead or Principal</b><br>8 to 12 years      | Architecture and multi team leadership.          | <b>45 to 80 LPA</b> |
| <b>Director or Head of AI</b><br>12 plus years | Strategy. Top product firms pay well above.      | <b>80 LPA plus</b>  |

*Figures are broad, indicative annual ranges in Indian Rupees and vary significantly with company, location, skills and global roles. Top product companies and overseas positions can pay considerably more.*

# Where the demand is

## Sectors

- IT and software, e commerce
- Fintech, healthcare, automotive
- Defence, agriculture, telecom
- Edtech, media and entertainment

## Organisations

- Global. Google, Microsoft, Amazon, NVIDIA, Meta
- Indian IT. TCS, Infosys, Wipro
- Startups. Sarvam AI, Krutrim and more
- Research. ISRO, DRDO, IITs, IISc

## Geographies

- Bengaluru, Hyderabad, Pune
- Gurugram, NCR, Chennai
- Global. US, Europe, Singapore, UAE
- Remote, freelance and consulting

# Recommended institutions

## Premier Engineering Institutes

- **IITs.** Bombay, Delhi, Madras, Kanpur, Kharagpur, Hyderabad, which has a dedicated AI department
- **IISc Bangalore** for leading AI research
- **NITs.** Trichy, Surathkal, Warangal
- **BITS Pilani.** Pilani, Goa, Hyderabad

## Strong CS and AI Programmes

- **IIITs.** Hyderabad, Delhi, Bangalore
- Delhi University, DTU and NSUT
- VIT, Manipal, Ashoka University
- Reputed state and private universities with AI or DS courses

# Building employability for the long run

## Courses and Certifications

- DeepLearning.AI specialisations by Andrew Ng
- Google, AWS and Azure ML certifications
- Stanford CS231n for vision, CS224n for NLP
- **Kaggle** competitions and Kaggle Learn

## Internships and Portfolio

- Research internships at IITs and IISc
- Google Summer of Code and industry internships
- Open source and a **GitHub project portfolio**
- Build things and share them publicly

## Future Outlook

- **Generative AI and agents** are reshaping every industry
- Government push through the **IndiaAI Mission**
- Rising focus on **responsible and ethical AI**
- One of the fastest growing, highest demand careers

IN SUMMARY

# Build the future, thoughtfully.

Start with strong mathematics and coding, choose a pathway that fits you, keep building projects, and never stop learning. AI rewards curiosity and persistence.

Thank you. Questions and discussion.